

## 6.4 Data-Collection Strategies

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔

CS XI

Chapter 6: Digital Literacy

Lecture 03 of 14

★ Golden

### 6.4 Data-Collection Strategies ★ Golden

#### LEARN IT

Digital literacy is not limited to finding information that **already exists** on the internet. You should also learn how to gather **original** (new) information yourself. A **data-collection strategy** is a planned method for gathering this information.

The five common strategies are:

Interviews

• Surveys

• Prototypes

• Observation

• Simulations

| Strategy    | How it works (one line)                               | Usual data type                              | Best when...                                      |
|-------------|-------------------------------------------------------|----------------------------------------------|---------------------------------------------------|
| Interview   | Planned conversation with a person or small group     | Mostly qualitative                           | You need deep, detailed answers from a few people |
| Survey      | Same questions asked to many people                   | Quantitative and qualitative                 | You need answers from many people quickly         |
| Prototype   | Users test an early version and give feedback         | Both                                         | You want to improve a product or solution         |
| Observation | Watch and record; no questions asked                  | Notes (qualitative) or counts (quantitative) | You want to see what people actually do           |
| Simulation  | Computer model; change variables and see what happens | Mostly quantitative                          | Real experiments are costly, risky or difficult   |

#### REMEMBER IT

Mnemonic for the five strategies: **"I S P O S"** – **"I Saw People On Screens"**: Interviews, Surveys, Prototypes, Observation, Simulations.

#### اردو میں سمجھیں

ڈیجیٹل خواندگی صرف موجود معلومات تلاش کرنے تک محدود نہیں، بلکہ نئی معلومات خود جمع کرنا بھی اس کا حصہ ہے۔ ڈیٹا جمع کرنے کے پانچ عام طریقے یہ ہیں: انٹرویو، سروے، پروٹوٹائپ، مشاہدہ (Observation) اور سمیولیشن (Simulation)۔

#### EXAM TIP

"List and explain the data-collection strategies" is a frequent long question. Name all **five**, give a one-sentence definition and **one example** for each, and say what type of data each gives. Detailed notes on each follow in the next two topics.

#### CHECK YOURSELF

1. Name the five data-collection strategies.
2. What does "original information" mean?

Answers: 1. Interviews, surveys, prototypes, observation, simulations. 2. New information that you gather yourself, rather than information that already exists.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

### MCQ 1. Data-collection strategies include:

- a. Interview, survey, prototype, observation, simulation
- b. Only NAND
- c. Only OSI ping
- d. Only K-maps

### MCQ 2. You collect data when:

- a. Existing sources are not enough
- b. You already have the full census always
- c. The question is undefined
- d. You only need a meme

### MCQ 3. A strategy must match:

- a. The question and the people
- b. The teacher's favourite chart only
- c. Agile always
- d. TCP window

### MCQ 4. Bias in collection:

- a. Systematically distorts answers
- b. Improves SD always
- c. Is required in exams
- d. Is a primary key

### MCQ 5. Ethics: people should:

- a. Know and consent
- b. Be filmed secretly always
- c. Hand over passwords
- d. Be named in viral posts

### MCQ 6. This is ★ because papers ask:

- a. Name + when to use + one limit of a method
- b. Only Python turtles
- c. Only ER
- d. Only Big O

## Extra short questions

1. List five strategies from the book.
2. One advantage of planning methods first.
3. One ethical rule.
4. What is a biased question?

## Extra long question

1. Choose methods for: (a) why students skip practicals (b) how a new ID-card prototype feels (c) traffic at the gate. Justify. ★ Golden.

## Extra answer key

**MCQs:** 1. A 2. A 3. A 4. A 5. A 6. A

### Short

1. Interview, survey, prototype, observation, simulation (and others in 6.4).
2. You do not collect the wrong type of data for your chart later.
3. Informed consent; anonymise students.
4. "Don't you agree our canteen is terrible?" — pushes the answer.

### Long

1. (a) interview+survey (b) prototype test (c) observation/count or simulation. Limits of each. Ethics sentence.